

# Introduction to mechanistic ecological niche models

## Day 1: Introduction

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Organized by:



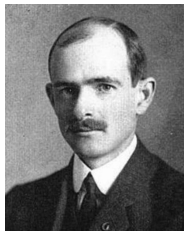
mechanistic ecological niche models

mechanistic ecological niche models

# What is the ecological niche?

## Grinnell (1917)

- niche as climatic and habitat requirements
- emphasis on abiotic conditions



## Elton (1927)

- niche as the role of a species
- emphasis on biotic interactions

These various circumstances, which emphasize dependence upon cover, and adaptation in physical structure and temperament thereto, go to demonstrate the nature of the ultimate associational niche occupied by the California Thrasher. This is one of the minor niches which with their occupants all together make up the chaparral association. It is, of course, axiomatic that no two species regularly established in a single fauna have precisely the same niche relationships.

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## Hutchinson (1957)

- niche as an **n-dimensional hypervolume**
- defined in environmental space
- conditions allowing population persistence

*"The niche is defined as an n-dimensional hypervolume"*  
— **Hutchinson (1957)**

# BAM diagram

Geographical space

G

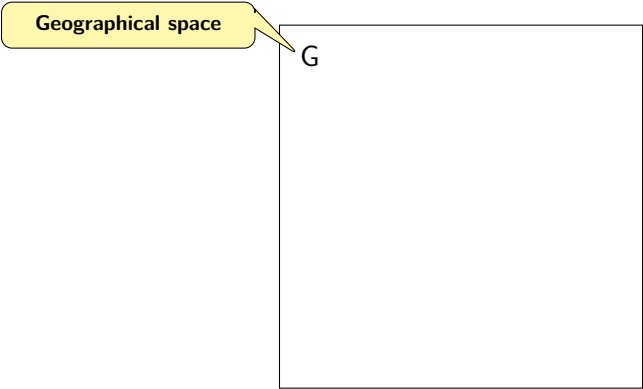


Figure 1: Based on Soberón and Nakamura (2009) and Sillero (2011).

# BAM diagram

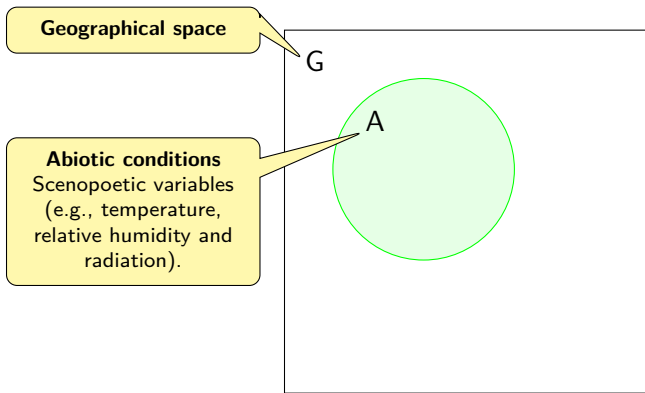


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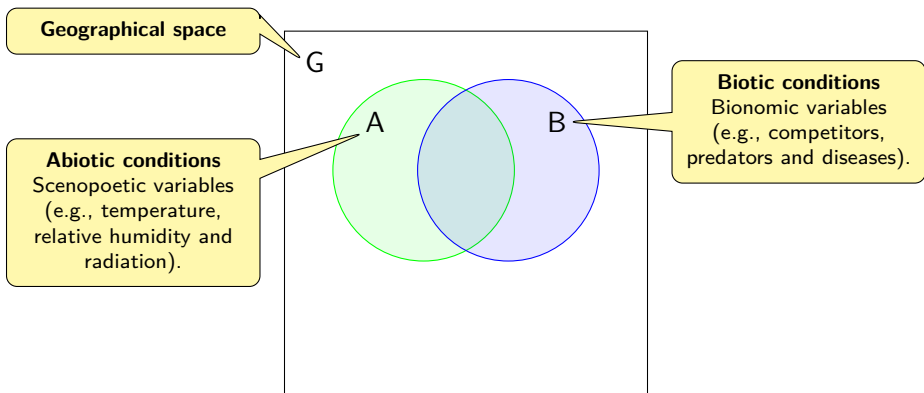


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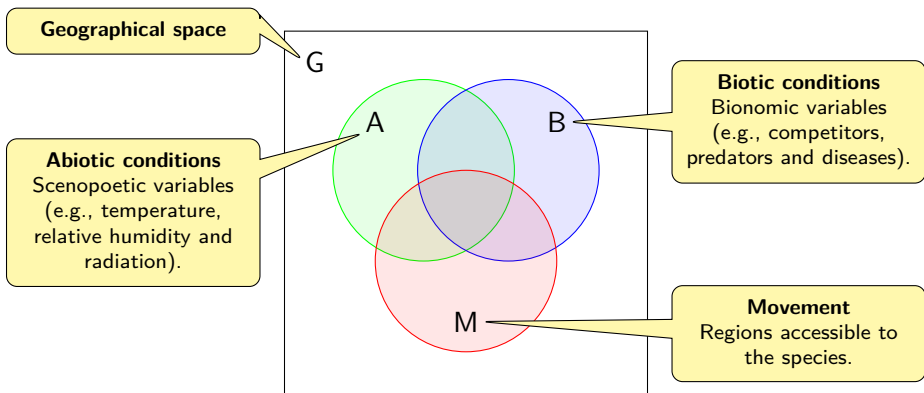


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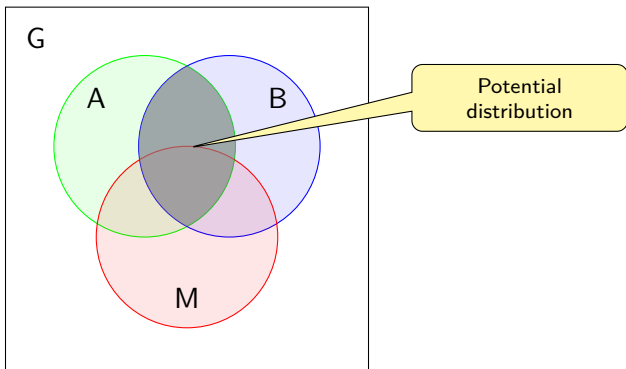


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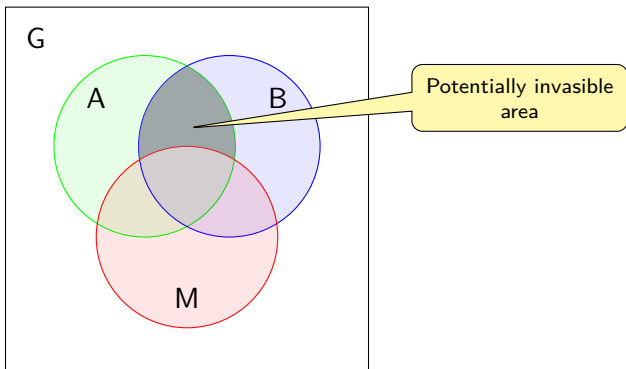


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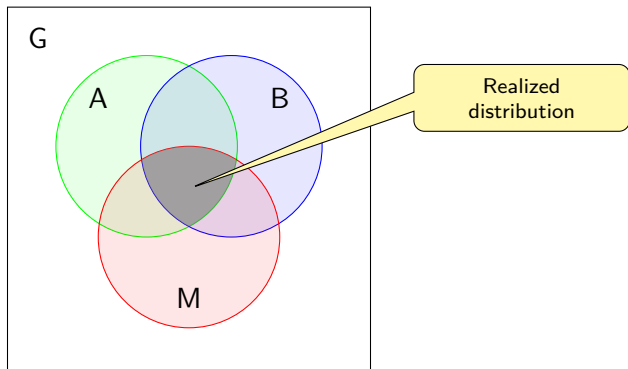


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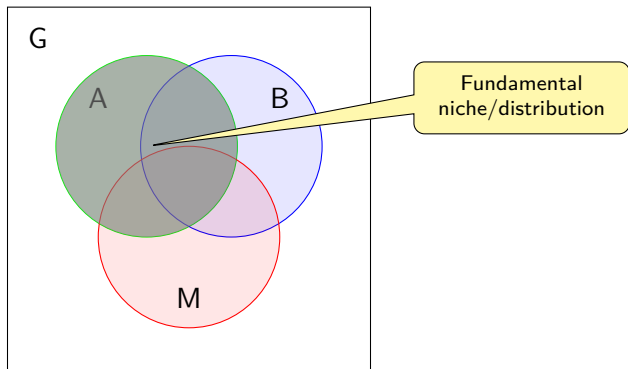


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# Mechanistic Ecological Niche Models

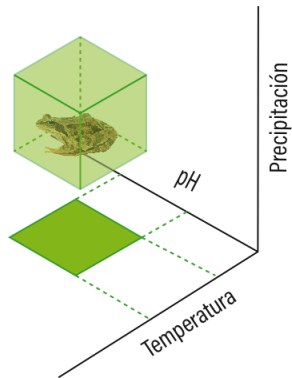


Figure 2: Taken from Maino et al. (2016)

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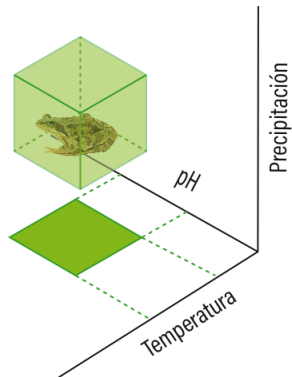


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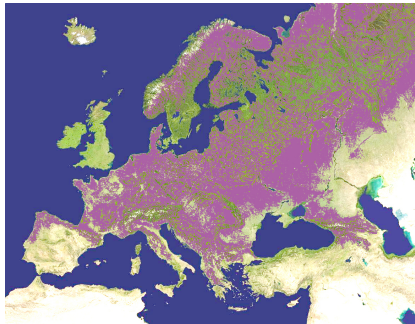
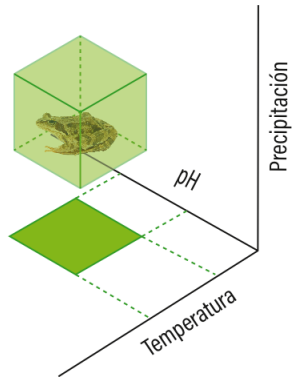


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mechanistic ecological niche **models**

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| predictive   | descriptive      |
| process      | pattern          |
| mathematical | statistical      |
| mechanistic  | phenomenological |

Modified from Bolker, [2008](#)

mechanistic ecological niche models

Mechanistic models are mathematical or computational representations that describe the underlying mechanisms of a system and its behavior by capturing the interactions and relationships between its components.

# The Lotka-Volterra Model

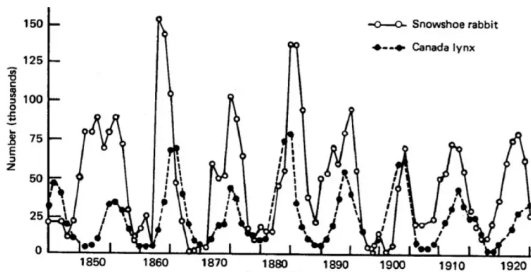


Figure 3: Population dynamics of rabbits and lynx

# The Lotka-Volterra Model

$$\begin{cases} \frac{dx}{dt} = \alpha x - \beta xy \\ \frac{dy}{dt} = \delta xy - \gamma y \end{cases}$$

## Parameters:

$x$ : density of prey (e.g., rabbits)

$y$ : density of predators (e.g., foxes)

$\alpha$ : intrinsic growth rate of prey

$\beta$ : predation rate

$\delta$ : predator reproduction rate

$\gamma$ : predator mortality rate

# The Lotka-Volterra Model

```
lv_model <- function(t, state, parameters) {  
  with(as.list(c(state, parameters)), {  
    dx <- alpha * x - beta * x * y  
    dy <- delta * x * y - gamma * y  
    list(c(dx, dy))  
  })  
}  
  
parameters <- c(  
  alpha = 0.6,  
  beta = 0.07,  
  delta = 0.04,  
  gamma = 0.9  
)  
  
state <- c(  
  x = lynxhare_df$Hare[1],  
  y = lynxhare_df$Lynx[1]  
)  
  
times <- seq(1, nrow(lynxhare_df), by = 1)  
  
out <- ode(  
  y = state,  
  times = times,  
  func = lv_model,  
  parms = parameters  
)
```

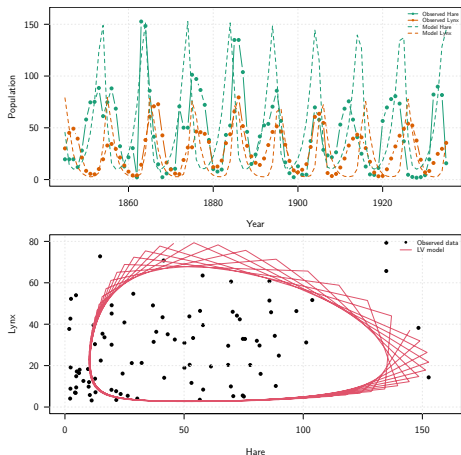


Figure 4: Observed hare-lynx dynamics and Lotka-Volterra simulation.

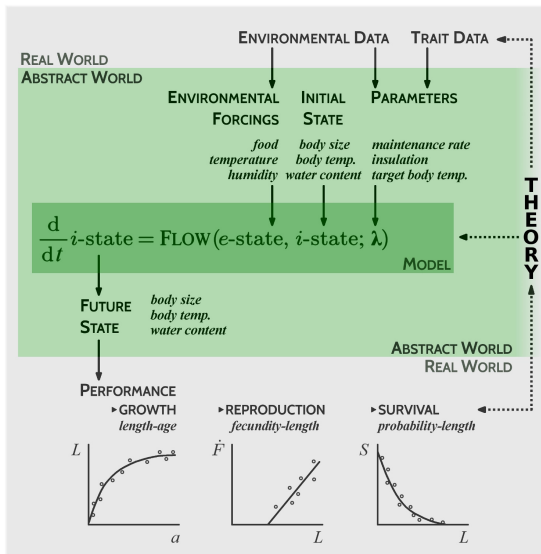


Figure 5: models and theory... Taken from Kearney et al. (2021)

mechanistic ecological niche models

# Species Distribution Models (SDMs)



Environmental  
layers

Figure 6: caption

# Species Distribution Models (SDMs)

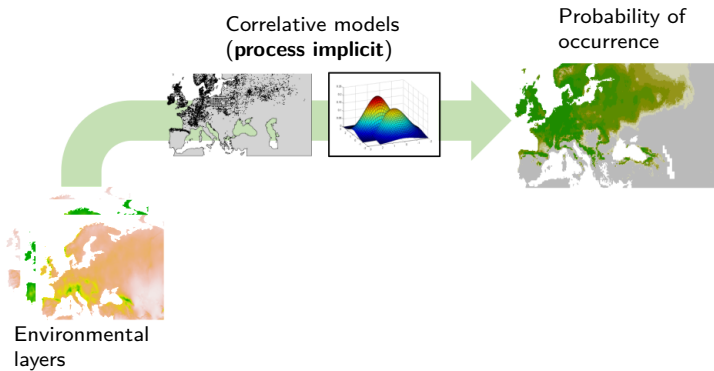


Figure 6: caption

# Species Distribution Models (SDMs)

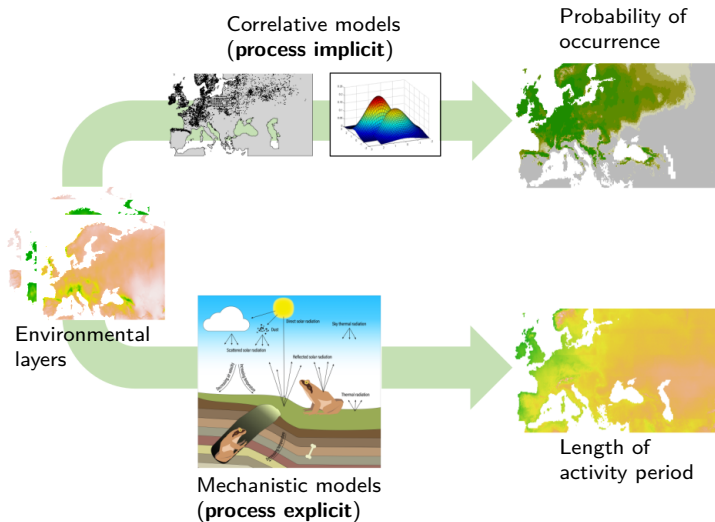


Figure 6: caption

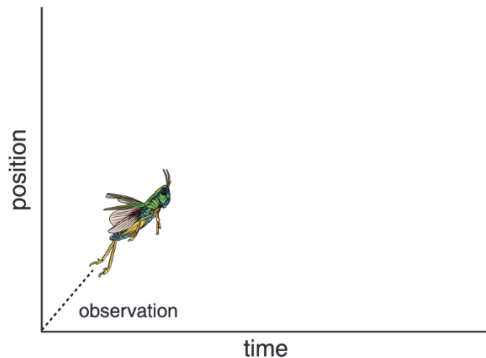


Figure 7: Taken from Maino et al. (2016)

# extrapolation

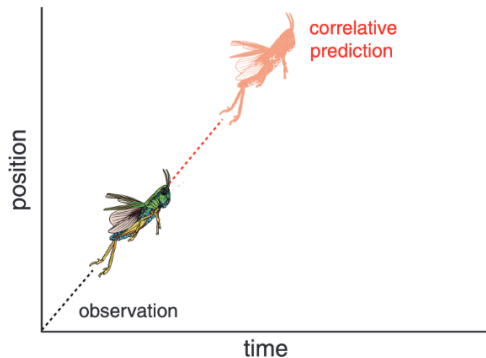


Figure 7: Taken from Maino et al. (2016)

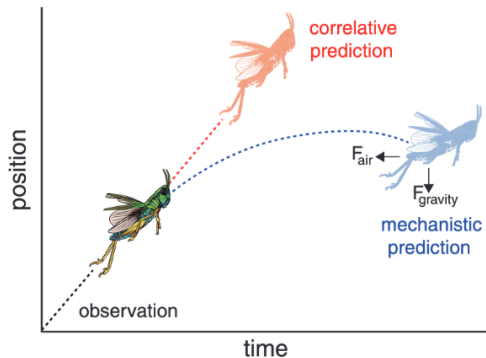


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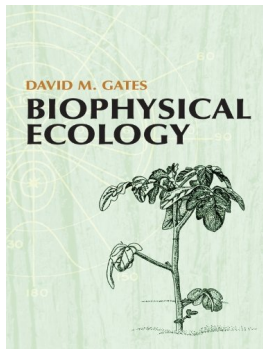
# Standing on the shoulder of giants



David M. Gates



Warren P. Porter



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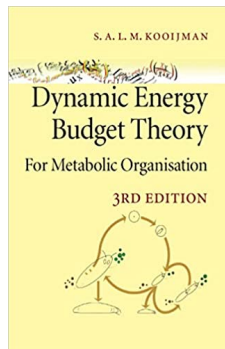
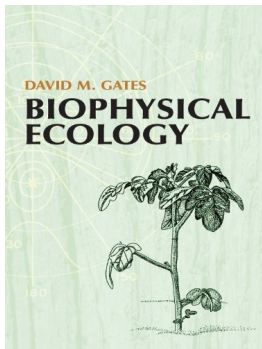
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Bas Kooijman



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